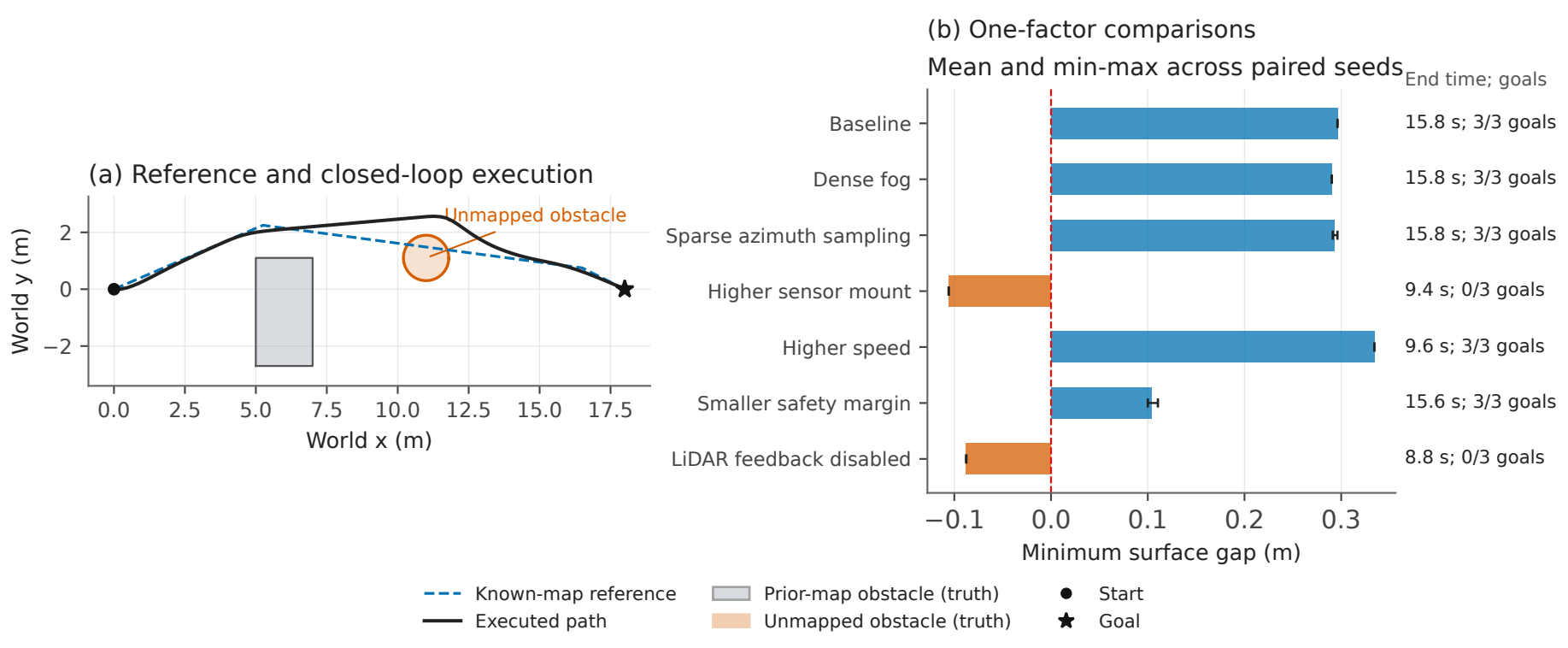
(a) Reference and closed-loop execution
Unmapped obstacle
World y (m)
2
0
−2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
World x (m)
(b) One-factor comparisons
Mean and min-max across paired seeds
End time; goals
Baseline
Dense fog
Sparse azimuth sampling
Higher sensor mount
Higher speed
Smaller safety margin
LiDAR feedback disabled
15.8 s; 3/3 goals
15.8 s; 3/3 goals
15.8 s; 3/3 goals
9.4 s; 0/3 goals
9.6 s; 3/3 goals
15.6 s; 3/3 goals
8.8 s; 0/3 goals
−0.1
0.0
0.1
0.2
0.3
Minimum surface gap (m)
Known-map reference
Executed path
Prior-map obstacle (truth)
Unmapped obstacle (truth)
Start
Goal